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BAGONG PILIPINAS

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SUBJECT: INFORMATION MANAGEMENT I

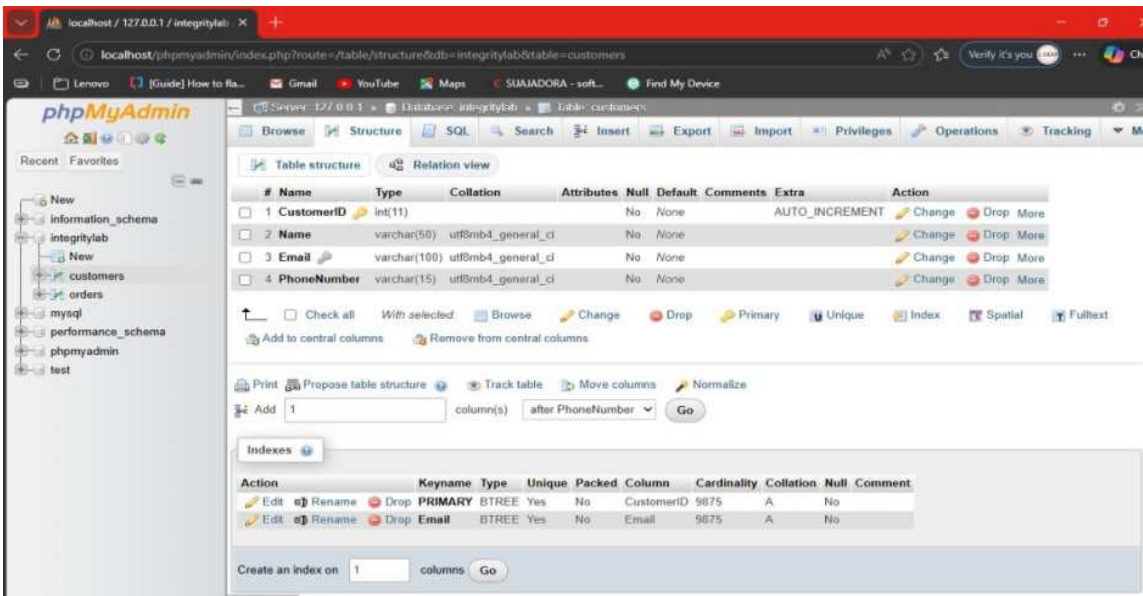
COURSE, YEAR & SECTION: BSIS 2

PROFESSOR: DR. GUILLERMO V. RED, JR.

Laboratory for Week 6

● **Customers Table (with additional constraint)**

```
CREATE TABLE Customers (  
CustomerID INT AUTO_INCREMENT PRIMARY KEY,  
Name VARCHAR(50) NOT NULL,  
Email VARCHAR(100) NOT NULL UNIQUE,  
PhoneNumber VARCHAR(15) NOT NULL  
);
```



● **Orders Table (with additional constraint)**

```
CREATE TABLE Orders (  
OrderID INT AUTO_INCREMENT PRIMARY KEY,  
CustomerID INT NOT NULL,  
OrderDate DATE NOT NULL,  
TotalAmount DECIMAL(10,2) NOT NULL DEFAULT 0.00,  
FOREIGN KEY (CustomerID)  
REFERENCES Customers(CustomerID)  
ON DELETE CASCADE  
);
```



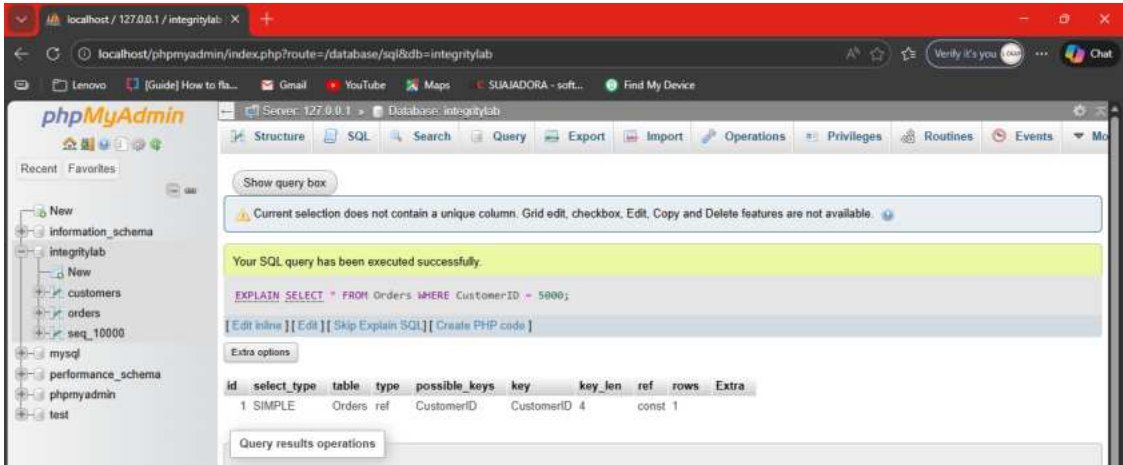


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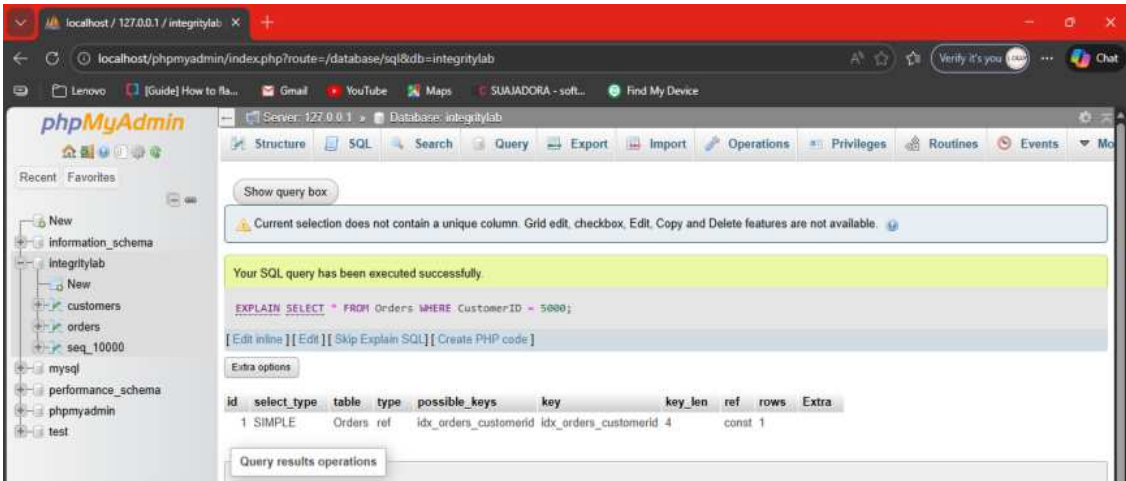
● **Indexing Performance (Before EXPLAIN)**

EXPLAIN SELECT * FROM Orders WHERE CustomerID = 5000;



● **Indexing Performance (After EXPLAIN)**

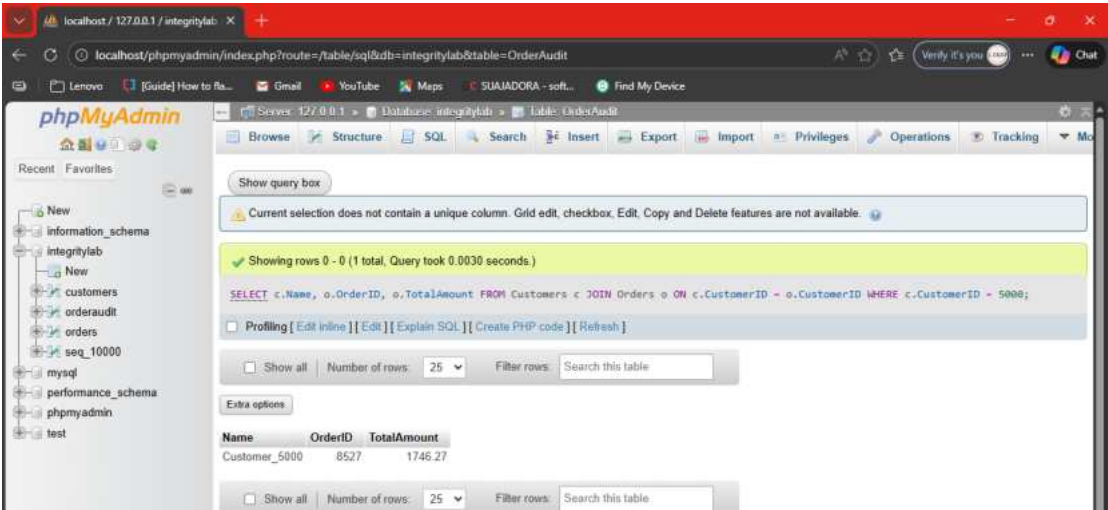
CREATE INDEX idx_orders_customerid
ON Orders(CustomerID);



● **1 JOIN Query**

SELECT c.Name, o.OrderID, o.TotalAmount
FROM Customers c
JOIN Orders o ON c.CustomerID = o.CustomerID
WHERE c.CustomerID = 5000;

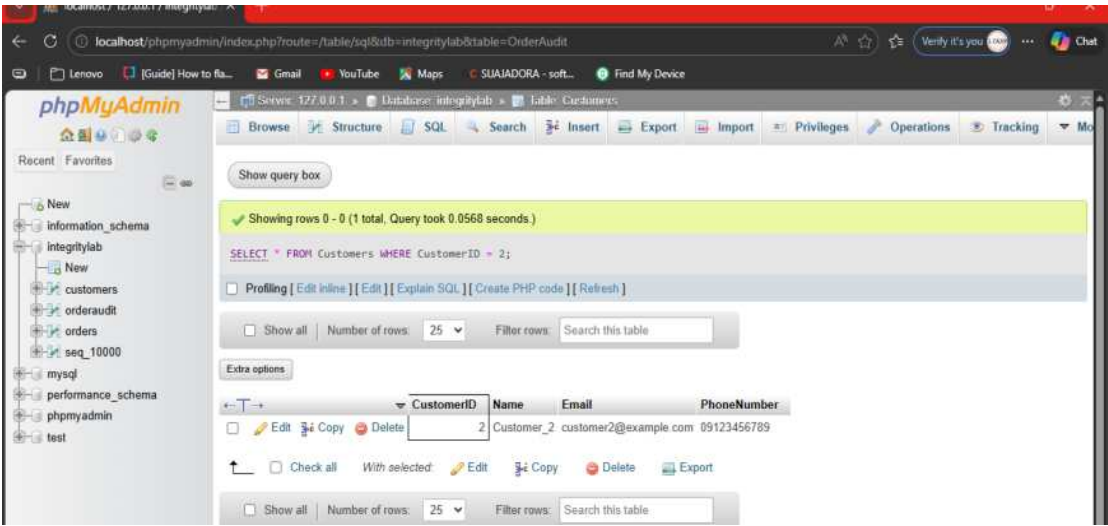




● **UPDATE Query Result**

UPDATE Customers
SET PhoneNumber = '09123456789'
WHERE CustomerID = 2;

SELECT * FROM Customers WHERE CustomerID = 2;



● **DELETE Query Result**

DELETE FROM Customers WHERE CustomerID = 3;

SELECT * FROM Customers WHERE CustomerID = 3;



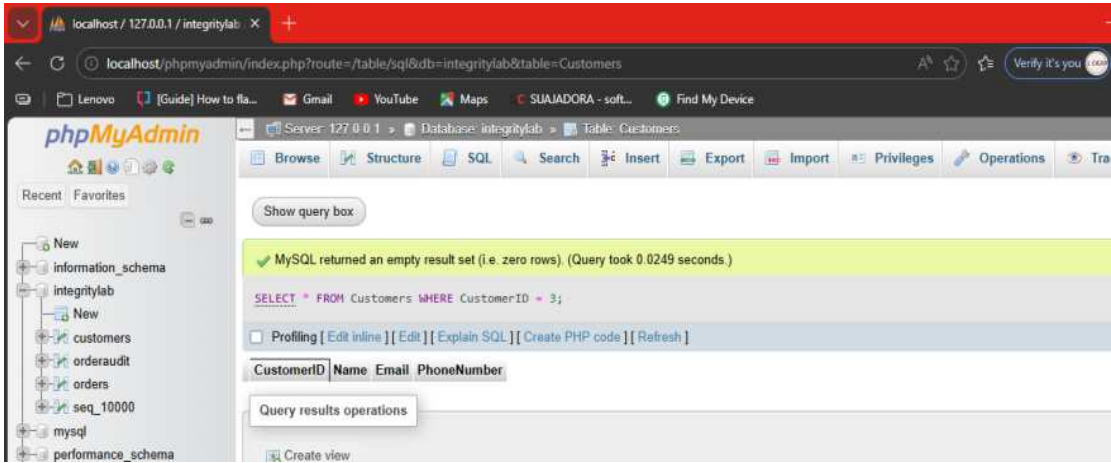


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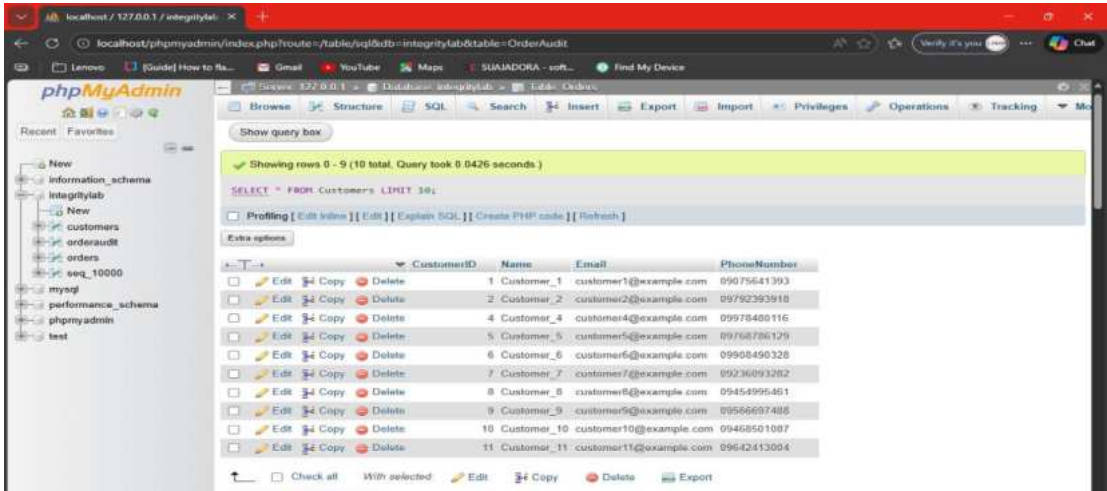


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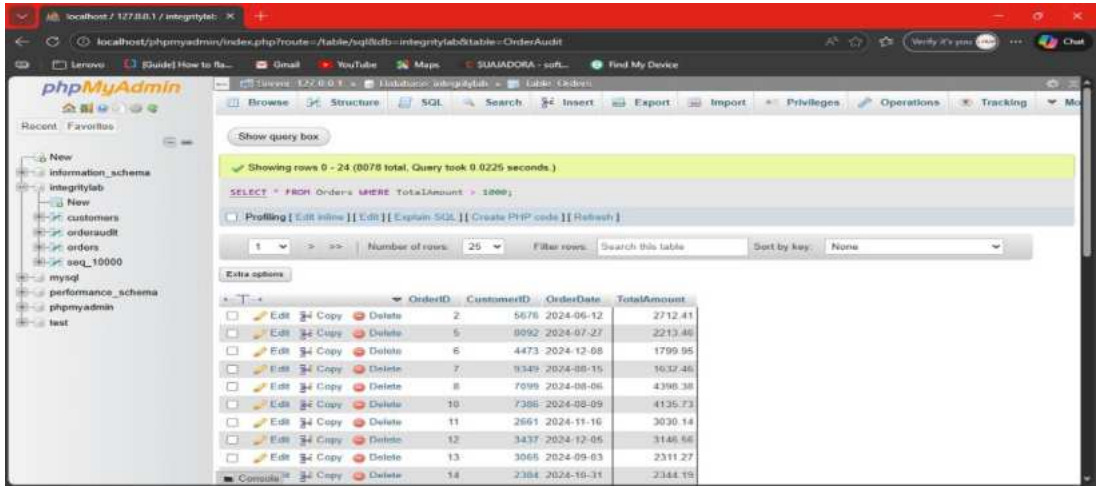


● 2 SELECT Queries

SELECT * FROM Customers LIMIT 10;



SELECT * FROM Orders WHERE TotalAmount > 1000;





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OrderID	CustomerID	OrderDate	TotalAmount
13	3055	2024-09-03	2311.27
14	2384	2024-10-31	2344.19
15	7900	2024-07-28	2530.86
16	7613	2024-04-25	1575.98
18	2532	2024-03-17	1470.10
19	7940	2024-02-13	1005.99
20	6826	2024-10-15	4586.60
21	1637	2024-02-05	5006.46
22	6671	2024-05-11	4078.35
23	9443	2024-04-21	3493.90
24	5310	2024-08-02	1796.57
25	9813	2024-11-09	1790.18
26	1651	2024-10-12	2107.31
27	7133	2024-05-01	2649.52
28	6037	2024-06-16	2464.13
29	378	2024-09-27	2993.27
31	7465	2024-07-25	2962.88

● **Test Transaction Rollback**

START TRANSACTION; UPDATE Orders SET TotalAmount = TotalAmount + 100 WHERE CustomerID = 5000; ROLLBACK; SELECT * FROM Orders WHERE CustomerID = 5000;	START TRANSACTION; UPDATE Orders SET TotalAmount = TotalAmount + 500 WHERE CustomerID = 5000; COMMIT;
--	---

OrderID	CustomerID	OrderDate	TotalAmount
543	5000	2024-10-30	3455.70
6790	5000	2024-02-18	998.48



● **Audit Log Trigger Test**

```
DELIMITER $$

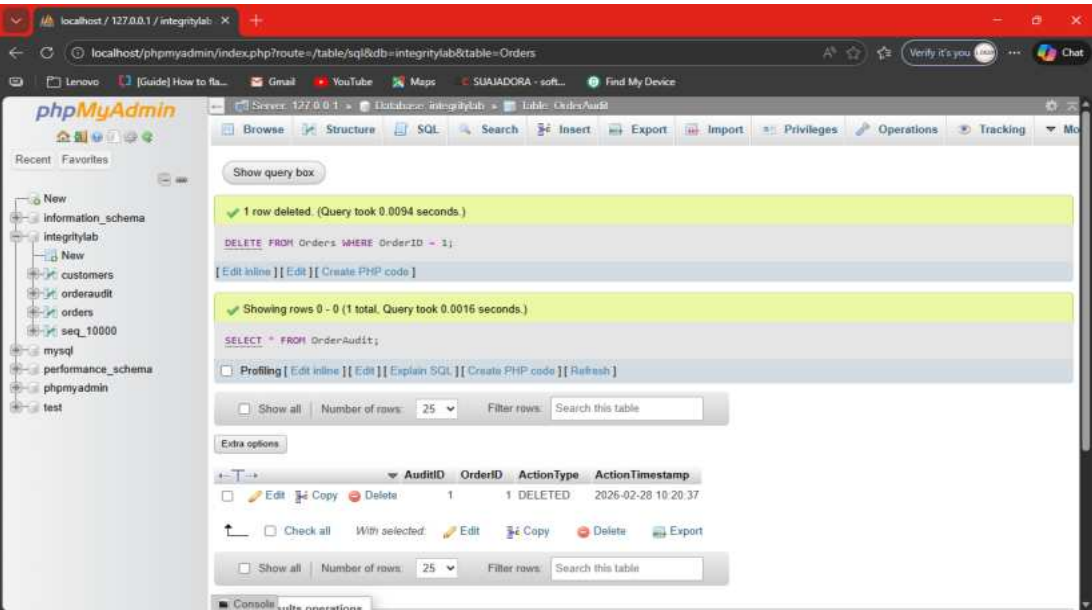
CREATE TRIGGER OrderDeleteLog
AFTER DELETE ON Orders
FOR EACH ROW
BEGIN
INSERT INTO OrderAudit (OrderID, ActionType)
VALUES (OLD.OrderID, 'DELETED');
END$$

DELIMITER ;

-- Test Audit

DELETE FROM Orders WHERE OrderID = 1;

SELECT * FROM OrderAudit;
```



REFLECTION:

This laboratory activity helped me understand how database integrity and performance work in a real system. I learned how primary keys and foreign keys connect tables correctly. The relationship between Customers and Orders makes sure that every order belongs to a valid customer. This helps prevent errors and keeps the data accurate.

The NOT NULL, UNIQUE, and CHECK constraints are important because they control the type of data that can be saved in the database. For example, the phone number must have 11 characters, and the total amount must be greater than zero. These rules protect the system from wrong or invalid data.





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The transaction activity helped me understand the ACID properties. When I used ROLLBACK, the changes were canceled and the data returned to its original state. When I used COMMIT, the changes were saved permanently. This shows how transactions protect the database from incomplete or accidental changes.

The indexing part showed how performance improves. Before creating an index, the database scanned many rows to find the data. After creating the index, the query became faster and more efficient. This is very important in web systems because fast response time improves user experience.

The audit trigger is also helpful because it records deleted orders automatically. This adds security and accountability to the system. Overall, this lab helped me understand how to build a stable, secure, and optimized database system.

